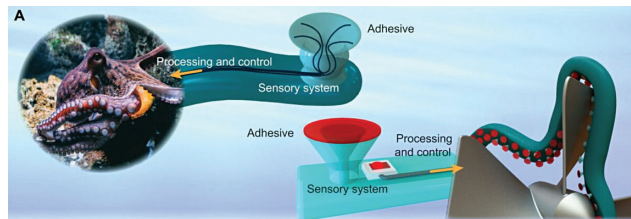


3D printing used to develop octopus-inspired gloves

Researchers from Virginia Tech in the US have used 3D printing to produce adhesive suckers that are inspired by octopuses! This was done by incorporating the 3D printing-enabled adhesive skin into a wearable glove. An assort-



Mimicking the nervous system of an octopus using electromechanical components (Credit: Virginia Tech)

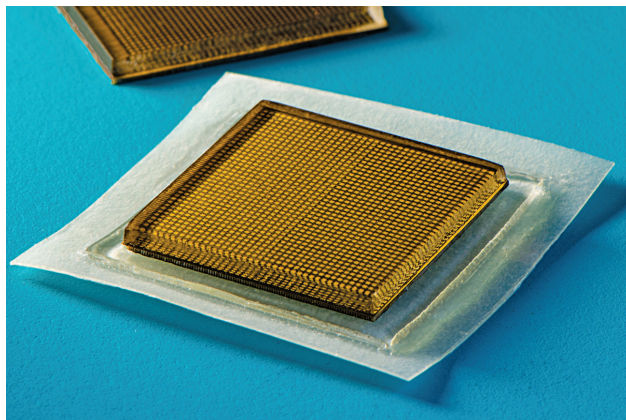
ment of lidar optical proximity sensors were built into each of the sticky suckers. Microcontrollers for real-time object identification were also included in the gloves. According to the research team, this hybrid of mechanical and electronic devices closely resembles the inner workings of the octopus nervous system. Although this study is focused on optical sensors, different sensing modalities could also be used in the future.

Machine language predictions made easy

Knowing when to accept an ML model's predictions is not always easy. Selective regression is a method that is usually used. However, it is not fool-proof—the likelihood that a model will make the correct prediction is not necessarily high for all subgroups. To address the issue, a team at MIT created two neural network algorithms that impose this fairness criterion. One approach ensures that all data on the sensitive qualities in the dataset, such as race and sex, is present in the features the model uses to make predictions. The second strategy makes use of a calibration technique to guarantee that the model consistently predicts the same value for an input, regardless of whether any sensitive attributes are added to that input.

High-resolution ultrasound imaging made convenient

MIT engineers have developed an imaging system to get high-resolution ultrasound images without causing inconvenience to the patients. This is a stamp-sized patch that consists of a rigid array of transducers and an adhesive layer that consists of hydrogel that's elastic and stretchy for high precision. This combination provides high-resolution

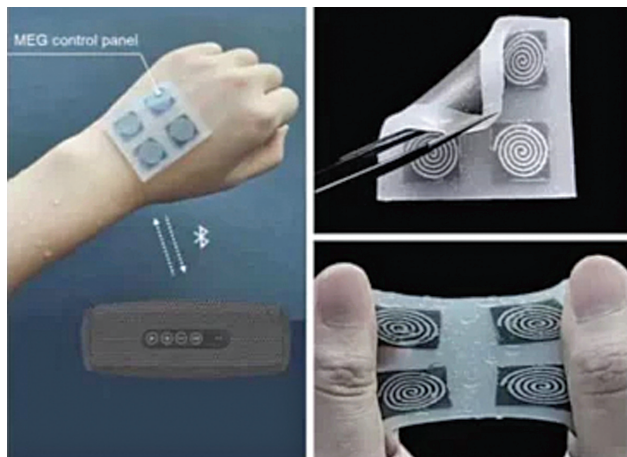


MIT engineers designed an adhesive patch that produces ultrasound images of the body. The stamp-sized device sticks to skin and can provide continuous ultrasound imaging of internal organs for 48 hours (Credit: <https://news.mit.edu>)

images of internal organs. The immediate goal of the researchers is to make this device wireless in such a way that patients can take this wearable imaging device home from a doctor or even buy it at a pharmacy.

A soft sensor to convert a finger press to electric signal

Scientists at UCLA have built a device using a soft magnetoelastic sensor array that is programmable, flexible, and durable and converts mechanical pressure to an electrical signal with a finger press. This wearable device is waterproof and



A programmable magnetoelastic sensor array for self-powered human-machine interface (Credit: Jing Xu et al, <https://doi.org/10.1063/5.0094289>)

overcomes the hurdles of a conventional human-machine interface (HMI) that stops functioning in the presence of human sweat. This HMI is used to operate electronic devices, such as music players and lamps. The researchers observed that a balance between performance and flexibility can be obtained by varying the thickness of the flexible material and the concentration of the magnetic particles.